

FH Aachen

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Expose

Evaluating AI Agent Capabilities in SysML v2 Modeling via a Semantic MCP Interface

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1 Expose

Model-Based Systems Engineering (MBSE) manages the growing complexity of cyber-physical systems through a central, formalized system model [1]. SysML is the established language for such models. Its most recent revision, SysML v2, rebuilds the language on the formally grounded Kernel Modeling Language (KerML) [2], with a textual notation and a standardized API for tool access.

This thesis empirically evaluates the SysML v2 modeling capabilities of LLM-based agents operating interactively on a live model inside the industrial tool Magic Systems of Systems Architect (MSoSA). A purpose-built Model Context Protocol (MCP) bridge connects the agent to MSoSA's Java Open API (plugin API) and exposes the live model state as tools. Based on the failures observed with this bare bridge, it is then extended by a harness of knowledge base, guardrails and encoded best practices intended to improve adherence to modeling conventions. The open-source Airbus Apollo 11 SysML v2 model [3] serves as the system under test, with the GfSE model collection [4] as a smaller secondary set.

In a recent review of LLMs for system modeling [5], interactive, tool-using agents on live models do not appear as an evaluated subject. The closest work either evaluates file-based tooling only [6] or contributes an MCP framework without a capability evaluation [7], and no standard benchmark for fundamental modeling tasks exists [8, 9]. Closing this gap matters, as industry deploys agentic MBSE tooling faster than it is measured.

RQ: *Which SysML v2 modeling tasks can an LLM-based agent perform reliably in an industrial modeling environment (MSoSA) when connected through an MCP bridge, and where do its capabilities break down?*

The evaluation is organized as a benchmark of six use cases on Apollo 11: **Create**, **Query**, **Patch** and **Delete** as primitives on the live model, plus **Verification** and **Validation**, distinguished according to ISO/IEC/IEEE 15288 [10]. Each use case is instantiated in several difficulty tiers, from single operations to composed, realistic modeling flows such as fault correction, with faults injected into the model following Cibrián et al. [11] and Al-Shami et al. [12]. The evaluation proceeds in two phases. First, the agent is benchmarked through the bare MCP bridge to establish a baseline. The collected failure data then drives the design of the harness, and the benchmark is repeated to measure the improvement. Every task has a ground truth, and the resulting model state is scored by an evaluator outside the agent loop [13]. Tasks are repeated to account for LLM non-determinism [14]. Runs use Claude-family models. The outcome is a quantitative capability profile per use case and tier, before and after the harness, plus a qualitative failure taxonomy [15].

Bibliography

- [1] INTERNATIONAL COUNCIL ON SYSTEMS ENGINEERING (INCOSE): Systems Engineering Vision 2020 / INCOSE. Version: September 2007. https://sdincose.org/wp-content/uploads/2011/12/SEVision2020_20071003_v2_03.pdf. 2007 (INCOSE-TP-2004-004-02, v2.03). – Technical Report. – INCOSE foresight report (workshop synthesis, not an empirical study) that coins the canonical MBSE definition. Anchors the thesis’s framing of MBSE and of the system model as the central artifact the agent operates on.
- [2] OBJECT MANAGEMENT GROUP: Kernel Modeling Language (KerML), Version 1.0 / Object Management Group. Version: March 2026. <https://www.omg.org/spec/KerML/1.0/PDF>. 2026 (formal/26-03-01). – Specification. – Normative OMG specification of the logic-grounded kernel SysML v2 extends, with four declarable tool conformance levels. Supplies the vocabulary for what an MCP bridge exposes – structural CRUD versus semantic conformance – and why v2 constraints are natively verifiable.
- [3] HELLE, Philipp ; SCHRAMM, Gerrit: Fly me to the Moon – Modeling Apollo 11 Using SysML v2. In: *Systems Engineering* (2026), June. <http://dx.doi.org/10.1002/sys.70074>. – DOI 10.1002/sys.70074. – [TOP 3] Airbus Central R&T’s five-layer CoSMA (Purpose/Operational/Functional/Logical/Technical) SysML v2 model of the Apollo 11 mission, released open-source (github.com/airbus/apollo-11-sysml-v2, MPL-2.0, commit 6e9c93f, accessed 20.09.2026) as a public MBSE benchmark. The instrument this thesis measures on: ~7.2 kLOC / 28 files / ~2 000 elements, deliberately incomplete “scaffold” for Create and fault correction tasks.
- [4] GESELLSCHAFT FÜR SYSTEMS ENGINEERING (GFSE): *SysML-v2-Models: An Open Collection of SysML v2 Models*. GitHub repository, <https://github.com/GfSE/SysML-v2-Models>, commit ebbb0c3, accessed 20.09.2026, 2025. – BSD-3. ~5.1 kLOC curated teaching/reference models (VehicleModel, DroneModelLogical, EveOnline Mining Frigate) explicitly meant to “help train LLMs”; ships `CI/parse.py` + a pilot-implementation jar as a validity oracle. Secondary set for generalisation: small self-contained models plus deliberate “bad examples”.
- [5] FRESEMANN, Carina ; ELLSEL, Claudius ; GRUNENWALD, Carl-Philipp ; STARK, Rainer: Using Large Language Models for System Modeling: Current State and Future Implications. In: *2025 IEEE International Symposium on Systems Engineering (ISSE)*, IEEE, October 2025, p. 1–8. – Structured review of 20 publications on LLMs for MBSE tasks (creation, analysis, modification, reformulation). Supplies the taxonomy behind this the-

sis's use-case catalogue, names MCP as the emerging context mechanism, and flags the missing evaluation criteria.

- [6] DEMPSEY, Colin: *Native AI Integration for Model-Based Systems Engineering: Three Layers that Make It Work*. SEI Blog, Carnegie Mellon University, <https://www.sei.cmu.edu/blog/native-ai-integration-for-model-based-systems-engineering-three-layers-that-make-it-work/>, September 2026. – [TOP 3] Controlled three-arm SysML v2 experiment (no tool / validation loop / plus MCP knowledge and skills): both tool arms reach zero syntax errors, but only full tooling lifts the pattern score (71.7 → 94.1). Direct template for this thesis's ablation arms; grey literature (practitioner blog), so its numbers are treated as unvalidated.
- [7] BAZZAL, Mahmoud ; SCHLEIFER, Simon ; LUNGU, Adriana ; KRUSE, Benjamin ; GOETZ, Stefan ; KOLDEWEY, Christian ; TISSEN, Denis ; WARTZACK, Sandro J. ; DUMITRESCU, Roman: Model Context Protocol for MBSE: A Unified Interoperable Agentic Framework for Systems Modelling. In: *Proceedings of the Design Society 6* (2026), July, p. 2721–2730. <http://dx.doi.org/10.1017/pds.2026.10630>. – DOI 10.1017/pds.2026.10630. – [TOP 3] MCP servers on the standard SysML v2 API with creation, modification and analysis tools, shown only through two qualitative case studies without metrics. Closest competing work: it delivers the artifact class, the thesis the missing capability evaluation inside MSoSA.
- [8] BOUAMRA, Yasmine ; YUN, Bruno ; POISSON, Alexandre ; ARMETTA, Frédéric: Sys-Temp: A Multi-Agent System for Template-Based Generation of SysML v2. In: *Advances in Practical Applications of Agents, Multi-Agent Systems, and Computational Social Science: The PAAMS Collection – 23rd International Conference, PAAMS 2025, Lille, France* Vol. 16031, Springer Nature Switzerland, June 2025 (Lecture Notes in Computer Science (LNAI)), p. 41–52. – Multi-agent template pipeline for NL-to-SysML v2 generation; scaffolding reaches zero syntax errors in 4/5 toy scenarios vs. 1/5 without, syntax only. The gap citation for this thesis's benchmark ("absence of standard benchmarks") and for SysML v2 corpus scarcity.
- [9] QUAST, Vincent ; JACOBS, Georg ; DEHN, Simon ; HÖPFNER, Gregor: Enabling Humans and AI Systems to Retrieve Information from System Architectures in Model-Based Systems Engineering. In: *Systems* 14 (2026), January, No. 1, p. 83. <http://dx.doi.org/10.3390/systems14010083>. – DOI 10.3390/systems14010083. – GraphRAG multi-agent QA over a Neo4j graph parsed from SysML v2 text, reaching 93% accuracy on a synthetic battery-EV model. Accuracy-metric template for Query and a second "no standardized benchmarks" gap citation; it queries a derived graph, not the live model.

- [10] ISO/IEC/IEEE: Systems and software engineering – System life cycle processes / International Organization for Standardization. Version: May 2023. <https://www.iso.org/standard/81702.html>. 2023 (ISO/IEC/IEEE 15288:2023). – International Standard. – Defines Verification (conformance to specified requirements) and Validation (fulfilment of stakeholder needs in the operational environment) as distinct life-cycle processes. The thesis adopts these definitions for its Verification and Validation use cases and audits prior work against them.
- [11] CIBRIÁN, Eduardo ; OLIVERT-ISERTE, Jose ; DíEZ-FENOY, Carlos ; MENDIETA, Roy ; LLORENS, Juan ; ÁLVAREZ-RODRÍGUEZ, Jose M.: Ensuring Semantic Consistency in SysML v2 Models Through Metamodel-Driven Validation. In: *IEEE Access* 13 (2025), p. 121444–121457. <http://dx.doi.org/10.1109/ACCESS.2025.3587786>. – DOI 10.1109/ACCESS.2025.3587786. – Metamodel-derived rule checking (lexical, ANTLR syntax, semantic cross-references) detects 97 % of 428 injected errors across 25 OMG test models. Despite its title, ISO-verification rather than validation; precedent for Verification: an error taxonomy, a fault-injection method transferable to fault correction, and a detection-rate baseline.
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- [13] WANG, Zizhe: *Beyond Executable Models: The Pufibara Agent Harness and the Modelica Agent Workflow Benchmark for Physical System Modeling*. arXiv:2608.23653 [cs.SE], <https://arxiv.org/abs/2608.23653>, August 2026. – Agent harness plus a source-grounded 232-task Modelica benchmark (repair, generation, tuning) scored by an evaluator outside the agent loop. Supplies three adopted principles: external evaluator, tasks grounded in an existing model, and the repair/generation/tuning split.
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peated runs. Template for the agent–tool division of labour and for repeated-run reliability reporting.

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- [17] AHLBRECHT, Alexander ; ZAESKE, Wanja ; DURAK, Umut: MBSqlE: Enabling SQL as Powerful Query Language for SysML v2 in Aviation. In: *44th AIAA DATC/IEEE Digital Avionics Systems Conference (DASC 2025)*. Montreal, Canada : IEEE, September 2025. – The standard SysML v2 API offers no graph traversal (REST traversals >10 s); MBSqlE imports models into SQLite so recursive SQL traverses them in one query. Motivates a semantic/cache layer instead of a thin REST proxy for Query.
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over three runs; few-shot examples prove essential. Primary reference for Create, but file-based without a modelling tool.

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- [24] LIPARULO, Leonardo ; PIERRI, Francesco: *Benchmarking AI Agents for Hardware Design Automation via MCP Tool Calling*. arXiv:2608.26199 [cs.AI], <https://arxiv.org/abs/2608.26199>, August 2026. – Benchmarks seven open-source models against a purpose-built MCP server for hardware design across single-operation,

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- [26] SHI, Jiasheng ; ZHANG, Tianhan: *What Do CAE Simulation Agents Really Need Beyond a Generic Harness?* arXiv:2609.03718 [cs.CE], <https://arxiv.org/abs/2609.03718>, September 2026. – With information access and repair budget held fixed, a generic single-agent harness beats specialised multi-agent CAE systems (96.4 % vs. 88.2 %); execution-feedback repair and domain tutorials drive the gains. Counter-evidence that the thin-vs-full-harness ablation must engage.
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